

A proof of Conjecture 00000006320

Two classes of 2×2 doubly stochastic matrices with the same permanent bound and different supports

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1 The conjecture

Conjecture 00000006320 reads (English version; the Chinese version agrees):

Definition: The permanent lower bound and the support structure are two layers. *Conjecture:* There exist two matrix classes with the same lower-bound constant but different support structures, and the separation is realized by an explicit pair with the same constant but different support families.

This is an existence claim. We give an explicit pair of matrix classes, prove that they have the same permanent lower-bound constant, and prove that their support families are different.

2 Reading and definitions

All matrices are real. The permanent of an $n \times n$ matrix is $\text{per } M = \sum_{\sigma \in S_n} \prod_i M_{\sigma(i),i}$ (Mathlib's `Matrix.permanent`).

Definition 1. A *matrix class* is a set $\mathcal{C}$ of real $n \times n$ matrices. Its *permanent lower-bound constant* $c(\mathcal{C})$ is the optimal constant c with $\text{per } M \geq c$ for all $M \in \mathcal{C}$, that is, $c(\mathcal{C}) = \inf\{\text{per } M : M \in \mathcal{C}\}$. This is the form of a permanent lower bound such as van der Waerden's $\text{per } M \geq n!/n^n$ for doubly stochastic matrices. We use that bound only as context, never in a proof.

Definition 2. The *support* of M is $\text{supp } M = \{(i, j) : M_{ij} \neq 0\}$. The *support family* (support structure) of a class is $\{\text{supp } M : M \in \mathcal{C}\}$.

The explicit pair. Let Ω_2 be the set of 2×2 doubly stochastic matrices (nonnegative entries, all row and column sums equal to 1; Mathlib's `doublyStochastic`). Let $\Omega_2^+ \subseteq \Omega_2$ be the subset of those with all entries positive.

3 The proof

Lemma 3. Every $M \in \Omega_2$ has the form $\begin{pmatrix} a & 1-a \\ 1-a & a \end{pmatrix}$ with $0 \leq a \leq 1$. Hence $\text{per } M = a^2 + (1-a)^2 \geq \frac{1}{2}$, with equality iff $a = \frac{1}{2}$.

Proof. The row sums give $M_{01} = 1 - M_{00}$ and $M_{10} = 1 - M_{11}$, and the first column sum gives $M_{10} = 1 - M_{00}$. So $M_{11} = M_{00} = a$. Nonnegativity of $M_{00} = a$ and $M_{01} = 1 - a$ gives $0 \leq a \leq 1$. Then $\text{per } M = M_{00}M_{11} + M_{10}M_{01} = a^2 + (1-a)^2 = \frac{1}{2} + 2(a - \frac{1}{2})^2$. $\square$

Theorem 4 (same constant). $c(\Omega_2) = c(\Omega_2^+) = \frac{1}{2} = 2!/2^2$. In both classes, $\frac{1}{2}$ is the least permanent: it is attained at $J_2/2$ (all entries $\frac{1}{2}$), which lies in Ω_2^+ .

Proof. Lemma 3 gives the lower bound on $\Omega_2 \supseteq \Omega_2^+$, and $\text{per}(J_2/2) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. $\square$

Theorem 5 (different support families). The support family of Ω_2^+ is $\{\text{full}\}$. The support family of Ω_2 is $\{\text{diagonal}, \text{anti-diagonal}, \text{full}\}$. In particular the identity $I_2 \in \Omega_2$ has diagonal support, which does not occur in Ω_2^+ .

Proof. Members of Ω_2^+ have all entries nonzero. In Ω_2 , Lemma 3 gives supports anti-diagonal for $a = 0$, diagonal for $a = 1$, and full for $0 < a < 1$. All three occur: at the swap matrix, at I_2 , and at $J_2/2$. $\square$

Theorems 4 and 5 together give the conjecture: (Ω_2, Ω_2^+) is an explicit pair of matrix classes with the same constant $\frac{1}{2}$ but different support families. $\square$

Remark 6 (scope and conventions). (1) We prove the constant in the strongest sense: an attained minimum (**IsLeast**) in both classes, and also as an infimum (**sInf**). (2) A support is a set of positions, not a set of positions up to row and column permutations. Even up to permutation, the support families differ: Ω_2 contains supports with 2 nonzero entries, while every support in Ω_2^+ has 4. (3) The example is the case $n = 2$ of the van der Waerden setting, with Ω_n^+ dense in Ω_n . We do not use the general van der Waerden theorem. (4) The reading of “explicit pair” is a pair of explicit matrix classes. The pair of explicit matrices $(J_2/2, I_2)$ is also exhibited: $J_2/2$ attains the common constant in both classes, and I_2 realizes a support that only Ω_2 has.

4 Lean formalization

File `lean/Conjecture6320/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1). It defines `lbConst` (infimum of `Matrix.permanent` over a class), `supp`, `suppFamily`, `Omega2 := Mathlib's doublyStochastic ℝ (Fin 2)`, and the positive subclass `Omega2pos`. The theorems are:

- Lemma 3, with $a = M_{00}$:
 - `Omega2_form`: the three entry identities;
 - `Omega2_bounds`: $0 \leq a \leq 1$;
 - `permanent_fin_two, perm_eq`: $\text{per } M = a^2 + (1 - a)^2$;
 - `perm_ge_half`: $\text{per } M \geq \frac{1}{2}$;
 - `perm_eq_half_iff`: $\text{per } M = \frac{1}{2}$ iff $a = \frac{1}{2}$.
- `isLeast_Omega2`, `isLeast_Omega2pos`, `lbConst_Omega2`, `lbConst_Omega2pos`, `const_eq_vdW`: Theorem 4.
- `suppFamily_Omega2`, `suppFamily_Omega2pos`: Theorem 5.
- `separation`: the main theorem. It states **IsLeast** for both classes, equal `lbConst`, `suppFamily_Omega2` $\neq$ `suppFamily_Omega2pos`, and that the support of I_2 lies in the first support family and not in the second.

lake build succeeds. `#print axioms` for `separation`, `suppFamily_Omega2` and `const_eq_vdW` reports only `[propext, Classical.choice, Quot.sound]`. The code contains no `sorry` and no `native_decide`.